

Hakan Tugrul Otal

PhD Student & Applied AI Researcher

University at Albany SUNY, New York, United States

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hakanotal.github.io — github.com/hakanotal — huggingface.co/hotal — Google Scholar

PhD Student and Research Assistant with a focus on quantifying and assessing risks of Generative AI systems. My research combines statistical modeling, ML, and cybersecurity to develop robust methodologies for AI risk evaluation and various applied AI projects. I bring hands-on experience in AI-driven risk engines for the insurance industry, published research on GenAI safety, and a strong foundation in experimental design and academic writing.

INTERESTS

AI Risk Assessment, GenAI Safety, Statistical Modeling, Cybersecurity, AI/ML, Large Language Models

EDUCATION

University at Albany, SUNY, New York, United States 08-2023 — Present

Ph.D. in Information Science

Data Analytics (Major) – Cybersecurity (Minor) GPA: 3.8/4.0

University at Albany, SUNY, New York, United States

M.S. in Information Science

Artificial Intelligence and Data Analytics GPA: 3.9/4.0

Istanbul Technical University, Istanbul, Turkey

B.S. in Computer Science

GPA: 3.5/4.0

EXPERIENCE

SUNY ORIED - Research Data Strategy

Summer Intern

06-2026 — Present

- Worked on SUNY Research Connect to support many campuses and researchers in SUNY.

University at Albany, SUNY

Graduate Research Assistant

08-2023 — Present

- Conducted research on Gen-AI security and safety at TATAR Lab under the supervision of Dr. Unal Tatar.
- Conducted research on integrating AI/LLM techniques for cyber threat detection at AI in Complex Systems Lab under the supervision of Dr. M. Abdullah Canbaz.
- Built and evaluated LLM-based pipelines and model outputs in collaboration with faculty and lab members across interdisciplinary projects.
- Supported various initiatives as a Graduate Assistant for Public Engagement and Enrollment.
- Presented research at various conferences and workshops.
- Teaching Assistant for CINF135: Concepts of AI.

CyRisk Inc.

AI/Software Engineer Intern

12-2025 — 05-2026

- Developed AI features for a cyber risk engine, including automated risk score calculation and analysis.
- Worked on cybersecurity litigation cases leveraging AI-driven insights for the tool used by cyber insurers and underwriters.
- Utilized API-driven development, deployment, and testing workflows with AWS, Docker, and Postman; coordinated with cross-functional teams on integration and issue resolution.

TATAR Lab (University at Albany, SUNY)

Summer Research Intern

05-2025 — 08-2025

- Worked on research projects related to AI security and safety.

- Worked on developing 2025 Cyber Insurance Report for NAIC (National Association of Insurance Commissioners) by analyzing/charting annual data about policies and coverages.
- Supervised student interns and led research group of undergraduate students.

AI in Complex Systems Lab (University at Albany, SUNY)

Summer Research Intern

05-2024 — 08-2024

- Worked on two research projects and published conference papers from them.
- Supervised undergraduate interns at UAlbany Summer Research Program (UASRP).

Parieta NLP

Software Engineer

01-2023 — 07-2023

- Designed and implemented frontend, backend, AI, and database for this legal tech startup.
- Developed the web application UI and integrated e-commerce features.
- Developed Nest.js backend services, MongoDB databases, and AI pipelines for the application.
- Deployed and maintained Docker containers on AWS infrastructure.

Deka Technology

Software Engineer

07-2022 — 12-2022

- Developed new API endpoints for enterprise-scale applications.
- Connected SAP services to backend systems through a customized integration layer.
- Worked with 100+ microservices using Jenkins, Openshift, Elastic, Redis, Quarkus, Apache Camel, and Keycloak.

PoiLabs

Software Engineer Intern

09-2021 — 10-2021

- Developed backend services for a mobile application for visually impaired people.
- Refactored the mobile application API with more dynamic and efficient approaches.
- Collaborated effectively using Bitbucket as VCS in a team setting.

PUBLICATIONS

- [1] Tyler H Merves, Michael H Conaway, Joseph M Escobar, Hakan T Otal, and Unal Tatar. **Systematic Capability Benchmarking of Frontier Large Language Models for Offensive Cyber Tasks**. In **2026 Systems and Information Engineering Design Symposium (SIEDS)**, pages 1–6. IEEE, 2026.
- [2] Hao Jian Huang, Hakan T Otal, and M Abdullah Canbaz. **Federated learning in adversarial environments: Testbed design and poisoning resilience in cybersecurity**. In **2025 IEEE International Conference on Communications Workshops (ICC Workshops)**, pages 1079–1084. IEEE, 2025.
- [3] Hakan T Otal, Abdulhamit Subasi, Furkan Kurt, M Abdullah Canbaz, and Yasin Uzun. **Analysis of gene regulatory networks from gene expression using graph neural networks**. In **Digital Healthcare, Digital Transformation and Citizen Empowerment in Asia-Pacific and Europe for a Healthier Society**, pages 249–270. Academic Press, 2025.
- [4] M Mikail Demir, Hakan T Otal, and M Abdullah Canbaz. **LegalGuardian: A Privacy-Preserving Framework for Secure Integration of Large Language Models in Legal Practice**. *arXiv preprint arXiv:2501.10915*, 2025.
- [5] M Abdullah Canbaz, Hakan Otal, Tugce Unlu, Nour Alhussein, and Brian Nussbaum. **Who Coordinates US Cyber Defense? A Co-authorship Network Analysis of Joint Cybersecurity Advisories (2024–2025)**. In *International Conference on Complex Networks and Their Applications*, pages 218–230. Springer Nature Switzerland Cham, 2025.
- [6] Hakan T Otal, Elyse Zavar, Sherri B Binder, Alex Greer, and M Abdullah Canbaz. **Harnessing Deep Learning and Satellite Imagery for Post-Buyout Land Cover Mapping**. In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 665–671. IEEE, 2024.

- [7] Hakan T Otal, Eric Stern, and M Abdullah Canbaz. **Llm-assisted crisis management: Building advanced llm platforms for effective emergency response and public collaboration.** In **2024 IEEE Conference on Artificial Intelligence (CAI)**, pages 851–859. IEEE, 2024.
- [8] Hakan T Otal and M Abdullah Canbaz. **Ai-powered crisis response: Streamlining emergency management with llms.** In **2024 IEEE World Forum on Public Safety Technology (WFPST)**, pages 104–107. IEEE, 2024.
- [9] Hakan T Otal and M Abdullah Canbaz. **Llm honeypot: Leveraging large language models as advanced interactive honeypot systems.** In **2024 IEEE Conference on Communications and Network Security (CNS)**, pages 1–6. IEEE, 2024.
- [10] Hakan T Otal, Stephen V Faraone, and M Abdullah Canbaz. **A New Perspective on ADHD Research: Knowledge Graph Construction with LLMs and Network Based Insights.** In **International Conference on Complex Networks and Their Applications**, pages 337–349. Springer Nature Switzerland Cham, 2024.

SELECTED PROJECTS

Autonomous Cyber-Offensive LLM Agents (VICEROY Project) 2026

- Leveraged Vertex AI to deploy large-scale, autonomous agents using a hybrid of frontier models and fine-tuned cybersecurity open-source models on Google Cloud infrastructure.
- Advanced the D-CIPHER multi-agent framework and contributed to the autonomous Cyber-Offensive LLM Agent research and red-teaming capability assessments of agents.
- Authored a research paper for presentation to Department of War (DoW) cyber spectrum leaders at the VICEROY Symposium.

NAIC’s 2025 Cyber Insurance Market Report 2025

- Contributed to the annual cyber insurance market analysis for the National Association of Insurance Commissioners (NAIC) by integrating and reconciling data from the Cybersecurity and Identity Theft Supplement and the International Insurers Department (IID).
- Synthesized multi-source regulatory and market data into statistical analyses and visual narratives on trends, policy coverages, and loss ratios for policy and industry audiences.

ATTASH - Anticipating Terrorist Technology Adoption to Secure the Homeland 2024

- Contributed to the ATTASH Tool Pilot as part of the development team.
- Developed Monte Carlo simulations in Excel and supporting Python scripts for statistical analysis for predicting terrorist technology adoption.

Distributed Network Load Generation Software 2023

- Bachelors Graduation Project

SERVICES

IEEE

IEEE Student Member 09-2023 — Present

IEEE Computational Intelligence Society (CIS) Member 03-2024 — Present

ACM

Student Member 10-2024 — Present

University at Albany, SUNY

Member of AI in Complex Systems Lab 08-2023 — Present

Member of TATAR Lab 05-2025 — Present

Graduate Assistant of Assistant Dean for Public Engagement and Enrollment Spring 2025

Supervisor of Student Interns and VICEROY Research Group at TATAR Lab Summer 2025

Supervisor of Undergrad. Interns at UAlbany Summer Research Program (UASRP) Summer 2024

Teaching Assistant for ‘CINF135: Concepts of AI’ Spring 2024, Fall 2024

Vice President, UAlbany Turkish Student Association 03-2026 — Present

AWARDS & GRANTS

University at Albany, SUNY

The Catalyst Award for Faculty Mentorship of Undergraduate Research, Scholarly Inquiry, and Creative Endeavors (Nominee)	2026
Student Travel Grant to attend RSAC 2026	(\$900) — 2026
Ph.D. Full Tuition Scholarship and Stipend	(\$32K) — 2023-2028

SUNY AI Platform – Google Cloud

Cloud computing grant via SUNY AI Platform (Google Cloud)	(\$30K) — 2026
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RISE Fellowship

Fellowship at Research and Innovators Startup Exchange (RISE) entrepreneurship program	(\$1290) — 2025
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IEEE Computational Intelligence Society

Student Travel Grant for IEEE CAI	(\$1200) — 2024
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IEEE Communications Society

Student Travel Grant for IEEE CNS	(\$800) — 2023
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WORKSHOPS & TALKS

Hot Topics in the Science of Security Symposium (HotSoS) Talk	2026
NTIR Workshop – “Building a Chatbot with RAG from Scratch”	2026
ACM UAlbany Workshop – “Deploying Models on the Edge”	2025

REVIEWED (TPC)

Conference on Computational Linguistics (COLING)	2025
Information Security Journal: A Global Perspective	2024, 2026
IEEE Conference on Communications and Network Security (CNS)	2024
IEEE Conference on Local Computer Networks (LCN)	2024
IEEE International Conference on Computer Communications (INFOCOM)	2024
Complex Networks and their Applications (CNA)	2023, 2024

Recommendation Letters Available upon Request:

Dr. M Abdullah Canbaz

Assistant Professor

Department of Information Sciences and Technology

University at Albany SUNY, NY, USA

E-mail: mcanbaz@albany.edu

Website: albany.edu/cehc/faculty/m-abdullah-canbaz

Dr. Unal Tatar

Associate Professor

Department of Cybersecurity

University at Albany SUNY, NY, USA

E-mail: utatar@albany.edu

Website: albany.edu/cehc/faculty/unal-tatar